

Hate Speech as a Proxy of Structural Demographic Phases: the case of Italy 1240-2020 and USA 1810-2020

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Abstract

This paper investigates the long-term historical dynamics of hate speech by analyzing a newly assembled corpus, HistoHate, through the lens of the Structural-Demographic Theory. Using validated annotation schemas, we exploit two Large Language Models to score four levels of toxic language in historical texts, we evaluate the annotation with human raters and perplexity, and then correlate these scores with the five phases of the secular cycles defined by the Structural Demographic Theory. Our findings reveal that, although there is no correlation between the two, the dynamics of the secular cycles and abusive language use are interdependent, but not aligned in time. In particular, our findings suggest that the intensity of toxic language tends to peak in later phases of the secular cycles, but not always in the same phase. This temporal misalignment suggests a complex interplay between societal shifts and linguistic expressions of conflict. This research highlights the promising synergy between Structural-Demographic Theory and Computational Linguistics, extending their application beyond traditional areas.

1 Introduction and Theoretical Framework

Is there a relation between the rise of societal conflict and hate speech? Looking back to the 2000s, hate speech was less pervasive than today. Early social media platforms like Friendfeed and Myspace focused on connecting with friends and colleagues, showcasing interests, gathering useful information [10], and offering outlets or support for stress relief [34]. However, by the late 2010s, the information landscape on social networks had changed drastically. The rise of fake news and misinformation contributed

to increased opinion polarization [21], fostering echo chambers where algorithmic filtering and interactions with like-minded individuals primarily expose users to reinforcing their existing beliefs [13]. In this environment, people who are uncomfortable with the prevailing beliefs may experience social anxiety or loneliness and struggle to find online support, negatively affecting their well-being [26].

Furthermore, these individuals have increasingly become targets of various forms of toxic language, including misogyny [6], hate speech directed at strangers [25, 14], attacks on political ideologies [20], and abuse targeting other vulnerable identities [19].

In particular, literature about abusive language [3] recognizes four levels that go from prejudice to violence:

1. abusive/toxic language, a broad category for prejudicial and harassing expressions (e.g: Migrants only know how to flaunt their idleness),
2. hate speech, that expresses hatred or discrimination even masked by irony (e.g: Nobody wants fake refugees to keep in hotels and we let dogs and pigs in),
3. aggressive language, that incites violence (e.g: Let’s hope these Roma with their toxic fires all burn and die quickly),
4. offensive language, that brings violence into action with pragmatics (i.e: Immigrants’ve pissed me off with all these terrorist acts. I am ready for war!).

All these levels can be used for addressing hate towards specific targets (i.e. sexist, homophobic, religious, etc.), and aggressive language can have an explicit intent to insult or offend other people. In this paper, we will use the terms “hate speech”, “abusive language” and “toxic language” interchangeably to refer the whole hate speech phenomenon.

The Structural Demographic Theory (SDT) [18] provides a macro-sociological framework for understanding the cyclical surge of hate speech. According to SDT, political instability is driven by the interaction of elite overproduction, declining popular living standards [24], and fiscal distress [30]. Using mathematical modeling and the SDT, researchers identified secular cycles in history, typically spanning 75 to 120 years [32]. SDT posits that three macro groups, the population, the elites and the State, operate within five distinct phases of these secular cycles (SDT phases):

1. The **growth** phase: marked by a novel and effective culture fostering social cohesion, this phase is characterized by increasing State control over the population [32]. The Elizabethan Age in England (1550s), the Roman principate under Augustus and post-war Italy (1950s) exemplify this phase.
2. The **population immiseration** phase: when the State’s capacity to control population reaches its limit, potentially causing rising inequality and social unrest between richer elites and poorer population. Examples of this phase are the Qing Dynasty in the early 19th century, the United States during the 1970s and Italy during the 1980s.

3. The **elite overproduction** phase: occurs when the population’s attempts to join the elite ranks overwhelm social mobility mechanisms. This leads frustrated elite aspirants to become counter-elites who challenge the status quo. Examples are university-educated radicals in the 1840s Russian Empire and lawyers in late 18th century France. Also Italy and United States in the 2000s are examples of this phase.
4. The **State stress** phase: marked by increasing fragmentation within the elite, which compromises the State’s ability to promote the cooperation between the population and elites. France in the 1780s is an example of this phase while Italy and the United States were in this phase during the 2010s.
5. The **crisis** phase: when the State can go towards collapse or recovery, depending on many factors. This phase culminates in a new social equilibrium and the commencement of a new period of stability, restarting the cycle. Examples include France in the 1790s, Italy in the 1940s, and the US during the Civil War in the 1860s and the New Deal era in the 1930s.

As the competition for limited elite positions intensifies and the "aspirant-to-position" ratio rises, intra-elite fragmentation often manifests as radicalized rhetoric used to mobilize popular discontent [31]. In this context, hate speech functions as a strategic tool for political entrepreneurship; it serves to dehumanize out-groups and solidify in-group identity, thereby accelerating the breakdown of social consensus. When demographic pressures such as a "youth bulge" coincide with stagnant social mobility, the resulting radicalization often turns into hate speech, as language is an expression of societies and cultures [33, 29]. This yields to the transformation of latent structural tensions into manifest social conflict [17].

1.1 Hypotheses, Method, and Limitations

The objective of this research is to investigate the macro-dynamics of toxic language across historical secular cycles. We formulate the following hypotheses to guide our analysis:

- Hypothesis 1 (H1): the progression through the five stages of the secular cycle is characterized by a growth of cultural tensions, marked by increasing social instability. As an expression of societies and cultures, language could reflect these mechanisms, and we should be able to observe an increase in toxic language in historical texts, corresponding to the secular cycles.
- Null hypothesis (H0): the relation between toxic language and the five phases of the secular cycle is randomic.

Method To test H1, we utilize a cross-national comparative approach focusing on Italy and the United States. Italy provides a deep diachronic perspective through its extensive literary and political history, while the United States offers large diachronic corpora, despite a shorter history. We map these texts by decade to the structural-demographic phase labels

provided by the Chronos Dataset [9]. Our methodology employs two Large Language Models (LLMs) to identify, categorize, and score toxic language within these historical texts. We calculate the mean toxicity scores per phase and perform correlation analyses to identify trends. To validate the LLMs’ efficacy on non-contemporary vocabulary, we compare model perplexity across modern and archaic texts. Furthermore, we assess the reliability of the automated hate speech scoring by benchmarking LLM annotations against human raters.

Limitations A primary limitation of this study is the imposition of contemporary moral and linguistic frameworks onto historical discourse. By utilizing modern definitions of toxic language, we apply our contemporary point of view on the past, as it is not possible to interpret the vision of hate speech formulated by past societies. Furthermore, we are aware of the survivorship bias of historical archives, the fact that texts of elites are more likely to be preserved than the informal or radicalized discourse of marginalized groups. This may result in an underrepresentation of the very counter-elite texts.

2 Data

We assembled a new corpus, named "HistoHate", from many different sources, from lyrics to newspapers and social media posts in Italian and English. This diversity is a deliberate feature of our research design. Structural-Demographic Theory explicitly differentiates between the popular and elite sectors of society, while the survivorship bias tends to preserve more elite texts. To capture the full spectrum of social tension, we do not want to rely solely on elite discourse (e.g., newspapers), but also on popular products, such as lyrics, social media and spoken dialogue transcripts, when available.

Our primary Italian source is CODIT¹ [23], a balanced diachronic corpus of written Italian containing roughly 29 million tokens from the 13th century to the 1930s. The text types within CODIT are diverse, ranging from poetry to fiction and scientific texts. In order to match the texts with SDT phases, we selectively sampled texts datable to a specific decade, excluding those with imprecise dating (e.g., "mid-14th century"). To bridge the gap from the 1940s to the 2020s, we incorporated random excerpts from other sources and corpora. For instance, we covered the 2020s using the Policycorpus [16], the 2010s with news from *corriere.it* [12] and data from *HaSpeeDe2018* [5], the 2000s with *Friendfeed* data [8] and the *LaRepubblica* corpus [4]. We covered the 1990s with *VOLIP* [2], a transcription of spoken Italian collected during that decade. For the period between the 1940s and 1980s, we sampled data from *L’Unità*, the Italian Communist Party’s newspaper. Additionally, we included randomly selected Italian song lyrics, play scripts, and some freely accessible portions of contemporary books. We also added historical texts that are

¹<https://www.korpus.cz/kontext/query?corpname=codit>

not present in CODIT, like Savonarola’s ”Trattato contro li astrologi” and Pico’s ”Libro detto strega”.

Our main American English data originates from COHA², the Corpus of Historical American English [1]. This corpus comprises approximately 385 million tokens spanning the 1810s to the 2000s, with each decade exhibiting a relatively consistent balance of fiction, popular magazines, news, and non-fiction books. To extend our coverage to the 2010s and 2020s, we incorporated social media data from WCPR13 [11] and WASSA-2021 [28], respectively. Similar to our approach with Italian, we also added randomly sampled American song lyrics from the 1960s to the 2020s.

The sources included in the HistoHate corpus feature heterogeneous categories. For instance, CODIT contains letters and play scripts, which are absent in COHA, while COHA includes magazine texts, not found in CODIT. To achieve homogeneity across the corpus, we standardized the text categories according to the mapping detailed in table 1.

Table 1: Mapping text types.

original types	mapping	%
newspapers	nws	30.7%
poetry, song lyrics	lyr	8%
social media	soc	0.4%
non-fiction, scientific texts	sci	4.1%
magazines, miscellaneous	mix	56.7%

The HistoHate corpus exhibits a high degree of structural heterogeneity, reflecting the evolution of available media and record-keeping from the 13th century to the present. Figure 1 illustrates the percentage of source distribution over time. The Italian sub-corpus transitions from a dominance of literary and poetic works (lyr, mix) in the medieval and Renaissance periods to a diverse modern assemblage of journalism (nws) and transcribed spoken or digital discourse (soc). The English sub-corpus mirrors this trend, showing a shift from 19th-century scientific and mixed prose to contemporary news and social media data.

To account for these disparate genre conventions and varying document lengths, our analytical framework does not rely on the raw frequency of toxic terminology. Instead, we utilize LLMs to calculate the average intensity of toxicity per decade. This approach normalizes the data by assessing the qualitative severity of abusive language within its specific linguistic context rather than its simple occurrence rate. By focusing on intensity, we mitigate the bias inherent in comparing high-volume, low-toxicity sources (such as scientific archives) with low-volume, high-impact radicalized texts. This ensures that the resulting toxicity scores reflect the level of systemic social tension regardless of the underlying medium of expression.

Many of the data sources have licenses that do not permit redistribution. In order to make HistoHate freely available , we extracted synthetic data with Gemini 2.0 flash.

²<https://www.english-corpora.org/coha/>

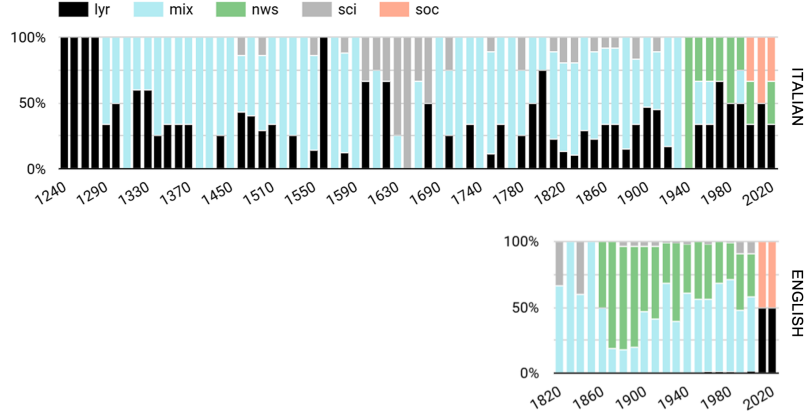


Figure 1: Types of texts in HistoHate by language and time.

Extract from the text the expressions that can be considered prejudicial, abusive, insulting, intimidating, harassing, and/or facilitate violence, hatred, or discrimination on the basis of race, ethnic origin, religion, gender, age, physical condition, disability, sexual orientation, political conviction, and similar. If there is no abusive language say "I could not find any abusive expression in this text."
Text: *DATA*

Figure 2: Prompt for the extraction of abusive language.

We selected Gemini for its input context window of ≈ 1 million tokens, and we asked it to extract the portions of text containing suspected abusive language. There is no consensus on a unique definition of abusive language [7], so we used a very broad definition [15] to design the prompt reported in Figure 2.

We split the input texts in order to fit the maximum size of 1 million tokens, we processed them with Gemini 2.0 flash and we generated the final text included in the HistoHate corpus. Due to the limits of Gemini’s output length, the portions of texts extracted cannot exceed 8192 tokens. These are some examples of the texts extracted, often followed by a synthetic analysis of the context and general evaluation of abusive language found, if any.

Ariosto: suppositi prosa.txt, 1500s, " *villano giudeo Turco Ebreo imbriaco*" [villain Jew, drunken Turkish Hebrew]

American magazine: 187781750.txt, 1870s, "black burglar good-for-nothing "

3 Annotation of SDT labels

We mapped the texts by decade to the SDT phase labels taken from the Chronos Dataset [9]. In order to have a replicable annotation method, the Chronos dataset included specific guidelines to interpret SDT phases from descriptions of historical decades, and reported the quality of the annotation with Fleiss' Kappa (κ) [Fleiss 1981], an inter-rater reliability metric. Achieving high agreement on this specific task is challenging due to the temporal nature of SDT phases. In practice, a single misalignment in phase labeling between two annotators can cascade through subsequent time-steps, causing the inter-annotator agreement to decrease exponentially. To mitigate this, The authors of the Chronos dataset introduced standards and constraints in the guidelines, improving the agreement from $\kappa=0.206$ to $\kappa=0.455$. The final replicable procedure for annotating the SDT phase labels involved two annotators, who were provided with descriptions of events happened in historical decades, and followed these guidelines:

- Read the textual description to identify key events: wars, reforms, rulers, population, elites, disasters, epidemics, alliances or treaties, socio-economic context, famines or financial stress, protests or movements, religions.
- Use polity identifiers to find the start and end decades of cultures. The end of a culture probably marks a crisis period.
- Starting from the beginning of a culture, initially assign the sequence of labels of a standard secular cycle model: 1,1,2,2,3,3,4,4,4,5 and then evaluate whether to keep or change the labels in each decade. It is possible to have longer or shorter cycles. There can be only one label 5 (crisis) per cycle. A polity can have one or more cycles.
- Having in mind the key events in the textual description, select one of the following labels to describe the decade: 1=growth. A society is generally poor when it experiences renewal or change followed by demographic (but not always territorial or economic) growth. Reforms, alliances, wars won or similar events are potential indicators of this phase. 2=impoverishment of the population. Potential economic and/or territorial expansion slows while demography continues to expand. The elite takes much of the wealth and defines the status symbols. Stability and external attacks are potential indicators of this phase. 3=Overproduction of the elites. The wealthy seek to translate their wealth into positions of authority and prestige. The population becomes poor. Movements, protests, and wars are potential indicators of this phase. 4=State stress. The elites want to institutionalize their advantages in the form of low taxes and privileges that lead the state into fiscal difficulties. Wars, protests and changes in the elite are potential indicators of this phase. 5=Crisis. a triggering event such as a war, revolt, famine or disaster that the state is unable to manage leads to a new configuration of society. Emigration of elites, subjugation to other societies, civil wars or profound reforms are potential indicators of this phase.

- Use the progressive order of the phases if no textual description is available for the decade.
- Make sure there is a progressive order of the labels (e.g. phase 3 must follow phase 2). All labels can be repeated in the following decade except the crisis phase, which conventionally lasts one decade.

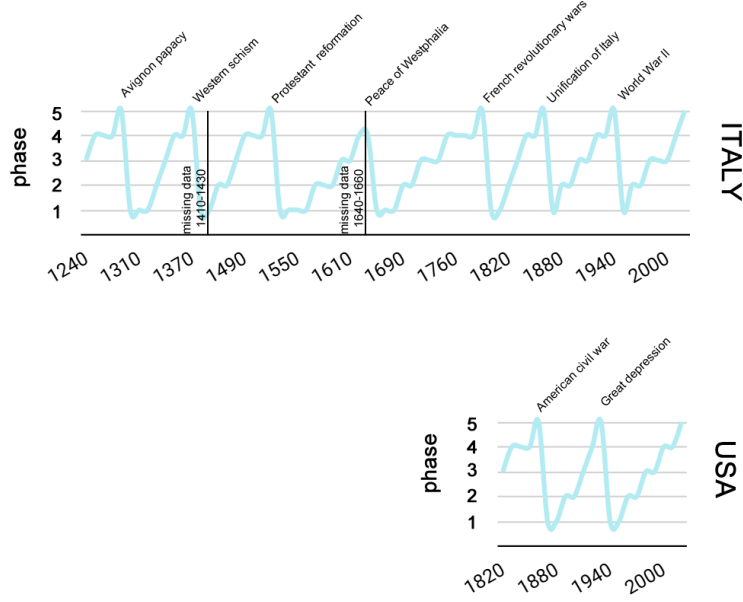


Figure 3: Distribution of SDT labels for Italy and United States.

The results of SDT phase annotation, mapped to Histohate data, are depicted in Figure 3 with the crisis periods annotated. The method shows robustness in identifying the chronological progression of SDT phases across both the Italian and American case studies. Apart from the gap around the Peace of Westphalia, when we do not have textual data available, As illustrated in the annotated data, the model correctly captures the theoretical sequence of phases 1 through 5, effectively mapping the transition from expansionary growth to systemic crisis. In the Italian timeline (1240–2020), Phase 5 peaks precisely align with major historical ruptures, including the Avignon Papacy, the Protestant Reformation, the Italian campaigns of the French revolutionary wars, and World War II. Similarly, the American data (1820–2020) identifies the sequential escalation toward Phase 5 during the American Civil War and the Great Depression. The successful identification of these cycles occurring in their predicted order validates the use of our linguistic and demographic proxies for tracking historical social stability.

4 Annotation of Hate Speech on Historical texts

To establish the validity of using LLMs for the analysis of toxic language, we first evaluated the models’ proficiency in processing historical texts like contemporary ones. We employed perplexity—a cross-entropy-based metric from information theory that quantifies a model’s uncertainty when predicting a sequence of text—as a proxy for linguistic comprehension. Since Gemini 2.0 flash is a closed model, it is not possible to compute perplexity directly on it, hence we computed perplexity on Gemma 7b, which is an open model based on the Gemini technology [22]. Results are reported in Figure 4

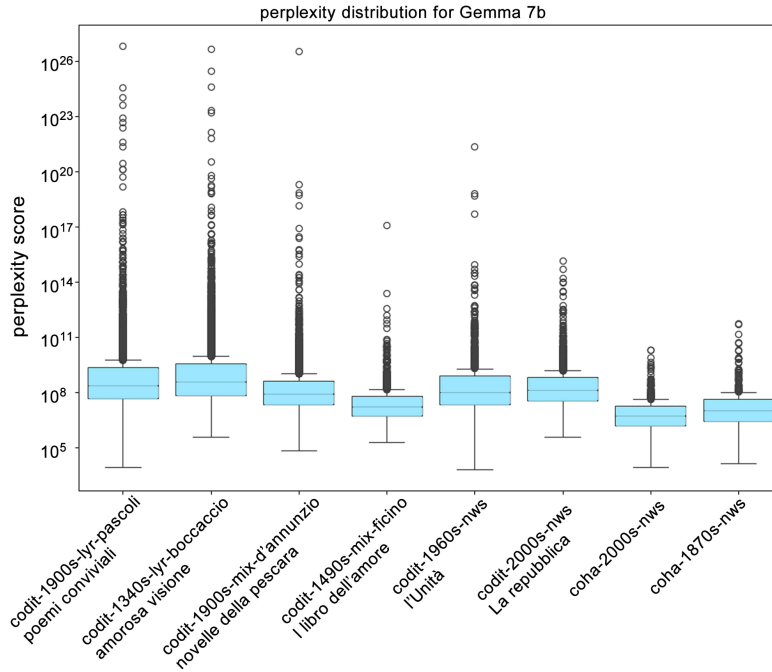


Figure 4: Perplexity of Gemma 7b on pairs of contemporary and historical texts.

As shown in the perplexity distribution for Gemma 7b, while historical texts (e.g., Boccaccio) naturally exhibit higher variance and slightly elevated median perplexity compared to modern news sources (e.g., La Repubblica), the scores remain within a comparable orders-of-magnitude range. This relative stability suggests that the LLM maintains sufficient performance to process historical variants of Italian and English, thereby justifying its use for extracting nuanced toxicity scores across different phases of the secular cycle.

By referring to existing hate speech annotation guidelines [Sanguinetti 2018], we engineered the following prompt to generate abusive language

Score expressions of hate in the text as: 0=no hate, 1=insults or judgments based on stereotypes. no incitement to hate; 2=dehumanizing offense without explicit incitement to hate; 3=explicit incitement to hate without taking part to the action; 4=explicit incitement to hate or violence with involvement in the action. Produce a json as output. Example input: "I could not find any abusive expression in this text", example output: {"abusiveTextScore":0}.
Text: *DATA*

Figure 5: Prompt for the generation of abusive language scores.

scores from texts in the HistoHate dataset with LLMs. In order to evaluate the hate speech annotation, we used two LLMs: we selected Gemini 2.0 flash for its demonstrated ability to process contemporary and ancient texts, and Claude 3.5 sonnet for its demonstrated effective moderation system to manage hate speech [Fasching 2025]. Figure 6 shows the distribution of the generated hate speech scores.

Figure 6 reports the distribution of the generated scores.

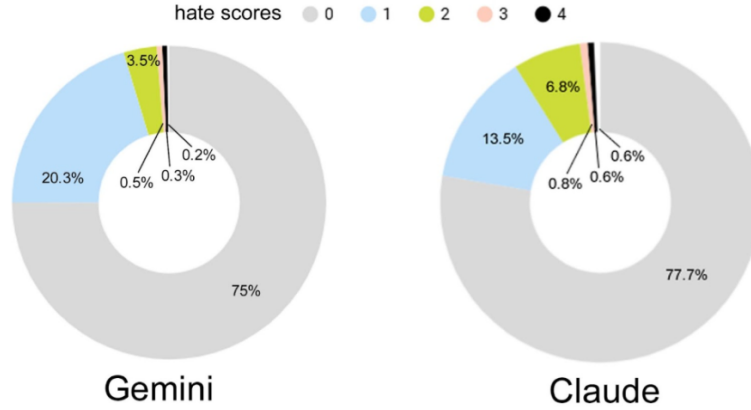


Figure 6: Differences in scoring abusive language with the two LLMs: Gemini 2.0 flash and Claude 3.5 sonnet. 0=no abusive language, 1=prejudice, 2=discriminatory expressions, 3=hateful expressions, 4=violent expressions. The white portion represents missing scores.

While both LLMs identify the majority of texts as non-abusive, they show some discrepancies in how they score levels 1 and 2. In general, Claude 3.5 Sonnet appears to assign lower hate speech scores than Gemini 2.0 Flash.

5 Experiments and Discussion

We computed the agreement between the two LLMs, the correlations and p-values between the abusive language scores and the SDT labels. In order to validate the LLM annotation, we used the same four-value schema [27], and asked three human raters to score abusive language in 98 examples, measuring alpha inter-annotator reliability [Krippendorff 2011].

Table 2: Evaluation of the data annotation. The table reports the experimental settings (Relation), examples (N), Krippendorff’s alpha agreement (α) and Pearson correlations (ρ). *=p-value below 0.05, **=p-value below 0.01.

Relation	N	ρ	α
Claude, Gemini	1405	0.85**	-
Claude, Claude run2	1405	0.96**	-
Gemini, Gemini run2	1405	0.99**	-
Claude, SDT labels	1405	0.02	-
Gemini, SDT labels	1405	0.05*	-
3 humans	98	-	0.091
Gemini, 3 humans	98	-	0.088
Claude, 3 humans	98	-	0.088
Claude, Gemini	98	-	0.392
Claude, Gemini 3 humans	98	-	0.118

We also ran correlation analysis filling the missing values with the most frequent score: zero. The results of the evaluation in Table 1 reveal that there is substantial agreement between the two LLMs. Moreover, generated scores are consistent, as we run the LLMs a second time, and the results have nearly perfect correlations to the original ones. In contrast, the agreement between annotators is really poor, given the difficulty of reading ancient texts, that sums to the subjectivity of the hate speech scoring task. Crucially, the LLMs have an inter-rater reliability much higher than humans in this task. The evaluation also reveals that there is almost no correlation between the generated abusive language scores and the SDT labels, but that the scores generated by Gemini are not casually distributed with respect to SDT phases. In fact, these scores have a p-value below 0.05 using the Pearson correlation, suggesting there is some non-casual linear connection between the rise of hate speech and the progression of structural-demographic phases.

The experiment does not support the full rejection of H0, and we designed a second experiment to understand why. We averaged the abusive language scores by SDT phase label in order to spot the values that are not in line with H1. In practice, we expect that the averaged abusive language scores increase with the labels.

Results, reported in Table 3, reveal that there is a linear increase in the abusive language scores generated by Gemini, while those generated by Claude have the average score of phase 1 (growth) that is higher than the average scores of phase 2 (population immiseration). This finding partially explains why the correlation of the scores generated by Claude have high p-value, but still remains the question why the correlation is

Table 3: Average abusive language score predicted by the two language models by SDT phase. Values not in line with H1 are marked in *italics*.

SDT phase	avg score Gemini	avg score Claude
1	0.243	0.294
2	0.260	0.266
3	0.335	0.358
4	0.379	0.378
5	0.483	0.492

generally so poor. Crucially, these results show that the dynamics of secular cycles and abusive language are linearly interdependent, but not necessarily aligned in time. To better understand this phenomenon we performed a qualitative analysis.

6 Qualitative Analysis

We plotted the result in a time chart, reported in Figure 7. The chart shows the dynamics of the secular cycles (SDTlabel) and the abusive language scores generated by Gemini and Claude, with peaks of toxic language indicated by black dots. In general, spikes of toxic language are found towards the end of secular cycles, in phases 3, 4 and 5, with the notable exception of the Western Schism, in particular around the 1380s. That decade in HistoHate is labelled as phase 1, while the crisis is assigned to the 1370s on the basis that the Schism started in 1378. If we consider this an annotation error, the consistency of peaks of instability and hate speech is even more robust.

In the Italian case there are 8 secular cycles, The first one (1180s-1290s) is marked by growing tensions between the Papal State and the Holy Roman Emperors, with nepotism and strong population control through religion, especially towards the final phases. This secular cycle is covered by textual data in HistoHate only since 1240s. In this period, Cavalcanti’s sonnets contain abusive language, specifically derogatory mockery of a woman’s physical appearance and threatening, intimidating rhetoric directed at political events. In addition, we find some mild prejudicial language based on regional/ethnic stereotyping (i.e: stereotyping of Genoese and Pisan people in Cenne della Chitarra). The following cycle, occurred roughly between 1300s and 1370s, is marked by the Avignon Papacy, when popes left Rome to seek for protection from the French kings, creating tensions between the Italian city-states siding with the Pope or with the Emperor. Here we find a spike of toxic language in phase 4, around 1350s. In particular, there are explicit hateful expressions and calls for violence against older women in Sacchetti’s “La battaglia delle belle donne”. The third secular cycle, between 1380s and 1520s, is characterized by the return of the pope in Rome and the expansion of the Papacy power, that brought wealth to the Guelph cities, sided with the pope. In this cycle we start to find expressions of hate against strangers from the earliest phases. In phase 2, around 1430s, there are insults against Bulgar-

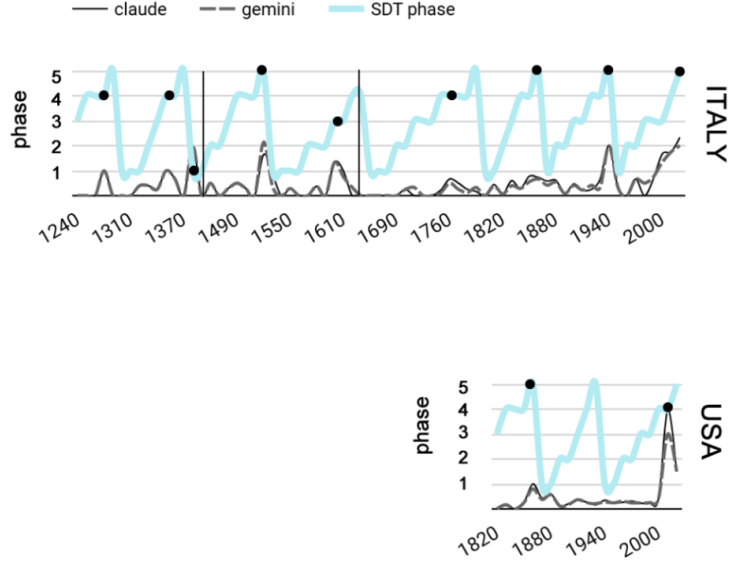


Figure 7: Dynamics of SDT labels through time and (averaged) abusive language scores generated by Gemini 2 flash and Claude 3.5 sonnet. Black dots indicate the SDT phases corresponding to spikes of abusive language.

ians in Pitti’s “Ricordi” and many occurrences of hate against Jews (e.g.: Castellani’s “Rimembranze”). In later phases, after 1500, hate speech against Jews and Moors can be found also in Ariosto. In phase 5 of this cycle, around 1520s, also returns hate against women, but in a stronger and violent form, exemplified by Giovanni Pico della Mirandola, which outlines procedures for torturing and executing accused witches.

The cycle between 1530s and 1650s is marked by the religious counter-reform, with a unity of interests between the elites (Italian nobles) and the State (Papal government) under the Spanish military protection. Unfortunately, there is no data for phase 5, possibly connected to emerging censorship practices in the Papal State since the 1530s, in response to the protestant reformation. However, this cycle has a spike in phase 3, around 1610s, when discriminatory expressions against Jews reappear in Marino’s “rime” and Andreini’s “Lo schiavetto”. The following cycle, between 1660s and 1780s, is characterized by the eclipse of papal influence in international affairs after the peace of Westphalia. Basically, this cycle is a long and slow decline of the Papal State and the rest of Italy. Here the two LLMs detected abusive language, mixed with irony and sarcasm, mocking poor people in Goldoni’s “Il campiello” (1750s), and growing toxic language after 1760s in Parini, Casanova, Pagano and other authors using discriminatory expressions - and insults - against women and aristocrats. From the secular cycle between 1790s and 1850s on, we have data for both Italy and the United States. While Italy is influenced by France,

the papacy being irrelevant in the international scenario, United States experience a fast growth (era of good feelings) followed by social conflicts about slavery. From the early phases of the cycle, around 1820s and 1830s, we find texts about "half-civilized Indians" in American magazines, and prejudice against strangers, Jews and gypsies in Manzoni, Leopardi and Romani. In the United States, toxic language spikes around 1860s, in phase 5 of the cycle, when we find a lot of derogatory and inflammatory language to demonize political opponents, while in Italy we find prejudicial expressions regarding religious and ethnic groups (Carducci, Guerrazzi). These tensions in language reflect the American Civil War between the Union and the Confederacy, and the emerging nationalism in Italy.

The cycle between 1870s and 1930s is characterized by reforms in the United States (progressive era), while the Kingdom of Italy saw WW1 and fascism. American newspapers and magazines contain prejudicial language based on ethnicity/race, specifically in reference to black people or other immigrants, while in Italy we can find body shaming, dehumanizing physical descriptions, misogynistic expressions and derogatory nicknames in authors like D'Annunzio and Benelli. The secular cycle between 1940s and 2020s has no spikes until phase 4, between 2010s and 2020s. Here social networks convey really high scores of hate speech against different targets such as strangers, politicians and women. The same targets of abusive and toxic language are also found in song lyrics, both in Italian and American English.

7 Conclusion

In conclusion, this study introduced a novel approach to investigating the historical macro-dynamics of hate speech by leveraging Computational Linguistics and the Structural-Demographic Theory. Our analysis of the HistoHate corpus revealed a pattern between the progression of secular cycles and the diffusion of abusive language. We demonstrate that LLMs maintain consistent linguistic performance across centuries, as evidenced by stable perplexity scores across diachronic corpora. The use of "intensity" rather than "frequency" for toxicity scoring effectively normalizes the varied source types, ranging from medieval lyrics to modern social media.

Our qualitative analysis confirms that peaks in toxic language intensity align specifically with phases 4 and 5 of the structural-demographic cycle. The sequential progression from Phases 1 through 5 is preserved in both regional models, validating the use of hate speech as a proxy of SDT. These initial findings offer compelling evidence for the potential to extend the applicability of the SDT framework into the realms of language and culture, opening new avenues for understanding the long-term interplay between societal structures and language.

Acknowledgments

Anonymized during the review process.

Ethical Statement

HistoHate combines historical data with LLM-annotated hate speech scores to help researchers understand and combat hate speech. We acknowledge the significant ethical considerations inherent identity in this work. The data is anonymized and released under a strict license to prevent misuse and protect individuals.

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